Computer Science - Undergraduate Level Assessment 3 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following is true of interrupts?
   (a) They are generated when memory cycles are "stolen".
   (b) They are used in place of data channels.
   (c) They can indicate completion of an I/O operation.
   (d) They cannot be generated by arithmetic operations.

2. Church's thesis equates the concept of "computable function" with those functions computable by, for example, Turing machines. Which of the following is true of Church's thesis?
   (a) It was first proven by Alan Turing.
   (b) It has not yet been proven, but finding a proof is a subject of active research.
   (c) It can never be proven.
   (d) It is now in doubt because of the advent of parallel computers.

3. Which of the following has the greatest potential for compromising a user's personal privacy?
   (a) A group of cookies stored by the user's Web browser
   (b) The Internet Protocol (IP) address of the user's computer
   (c) The user's e-mail address
   (d) The user's public key used for encryption

4. Let l = [1,2,3,4]. What is min(l) in Python 3?
   (a) 1
   (b) 2
   (c) 3
   (d) 4

5. Which of the following pairs of 8-bit, two's-complement numbers will result in overflow when the members of the pairs are added?
   (a) 11111111, 00000001
   (b) 00000001, 10000000
   (c) 11111111, 10000001
   (d) 10000001, 10101010

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the output of print(list[1:3]) for the list ['abcd', 786, 2.23, 'john', 70.2] is [786, 2.23].

7. True or False: The value of the Python expression 1 + 3 % 3 is 1.

8. True or False: Static allocation is sufficient for managing memory in recursive procedures due to their predictable memory usage.

9. True or False: The language { ww | w in (0 + 1)* } is accepted by some Turing machines but not by any pushdown automaton.

10. True or False: In Python 3, the function max(list) returns the item with the maximum value from a given list.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to caluclate the area of a tetrahedron.

12. Write a function to split the given string at uppercase letters by using regex.

End of Paper